

Short Term Frequency Domain Processing of Speech

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Some Good Books in Speech Processing

- L. R. Rabiner, B. H. Juang and B. Yegnanarayana, "Fundamentals of speech recognition", Pearson Education, 2009.
- L.R. Rabiner and R.W. Schafer, "Digital Processing of Speech Signals", Pearson Education, Delhi, India, 2004
- J. R. Deller, Jr., J. H. L. Hansen and J. G. Proakis, "Discrete-Time Processing of Speech Signals", Wiley-IEEE Press, NY, USA, 1999.
- D. O'Shaughnessy, "Speech Communications: Human and Machine", Second Edition, University Press, 2005.
- T. F. Quatieri, "Discrete time processing of speech signals", Pearson Education, 2005.



Organization

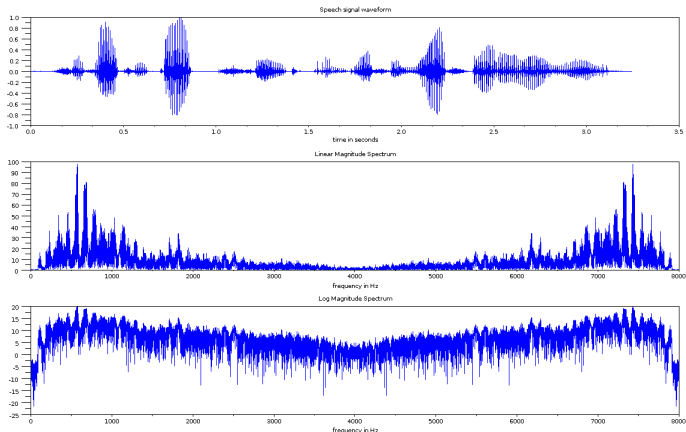
- Limitation of DTFT for spectral analysis of speech.
- Development of short-term Fourier transform (STFT) representation.
- Difference in the nature of linear and log magnitude spectra.
- Spectra of voiced, unvoiced and silence regions of speech.
- Plot STFT of a speech signal.
- Difference between true and convolved spectra.
- Effect of rectangular, Hamming and Hanning window functions
- Effect of frame size on short term spectral analysis

Need for Short Term Frequency Domain Analysis of Speech

- Frequency components present in speech are not directly apparent in time domain.
- Need to go for frequency domain representation using Fourier representation.
- Conventional Fourier representation studied in mathematics or signal processing is inadequate to provide information about time varying nature of spectral information present in speech.
- Hence need for short term version of Fourier transform, termed more commonly as Short term Fourier Transform (STFT)



Need for Short Term Freq Dom Processing



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Fourier Transform

- Frequency domain representation
- Continuous time Fourier Transform (CTFT)
- **CTFT** : $X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$
- **Inverse CTFT** : $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)e^{j\omega t} d\omega$
- Given CT aperiodic signal $x(t)$, we can do frequency domain analysis using CTFT pair of relations
- Signal $x(t)$ is assumed to be stationary
- Speech is a non-stationary signal and hence CTFT is not useful



CTFT of Periodic Signals

- Let $x(t) \leftrightarrow X(j\omega) = 2\pi\delta(\omega - \omega_0)$
- $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi\delta(\omega - \omega_0)e^{j\omega t}d\omega = e^{j\omega_0 t}$
- Multiplication of a function with an impulse
- Sampling/shifting property of an impulse
- Next, let $x(t) \leftrightarrow X(j\omega) = \sum_{k=-\infty}^{\infty} 2a_k\pi\delta(\omega - k\omega_0)$
- $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$
- Thus $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = X(j\omega) = \sum_{k=-\infty}^{\infty} 2a_k\pi\delta(\omega - k\omega_0)$
- LHS is CTFS coefs a_k based and RHS CTFT representation.
- Periodic signal $x(t)$ represented by CTFT as impulse train in FD, each impulse $(\omega - k\omega_0)$ weighted by respective Fourier coef a_k



CTFT of Impulse Train in TD

- Let $x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$ with period T
- $X(j\omega) = \sum_{k=-\infty}^{\infty} 2a_k \pi \delta(\omega - k\omega_0)$
- $a_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t} dt = \frac{1}{T}$
- $X(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{T})$
- CTFT of a periodic impulse train in TD with period T is a periodic impulse train in FD with period $\omega_0 = \frac{2\pi}{T}$
- Inverse relationship between time and frequency domains

Discrete Time Fourier Transform

- Frequency domain representation of DT signal $x(n)$
- Discrete time Fourier Transform (DTFT)
- **DTFT** : $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$
- **Inverse DTFT** : $x(n) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$
- Like CTFT, DTFT can also be used for both aperiodic and periodic signals or sequences
- DTFT also assumes $x(n)$ to be a stationary signal, but speech is a non-stationary signal
- Eventhough time domain signal $x(n)$ is discrete, its freq. dom. representation $X(e^{j\omega})$ is continuous and hence not suitable for digital processing in frequency domain



Short Term Fourier Transform

- Windowed segment of speech $x_w(n) = x(n)w(n - m)$
- **DTFT** : $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$
- **STFT** :
$$X_w(e^{j\omega, n}) = \sum_{m=-\infty}^{\infty} x_w(m)e^{-j\omega m} = \sum_{m=-\infty}^{\infty} (x(m)w(n - m))e^{-j\omega m}$$
- **Inverse DTFT** : $x(n) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$
- **Inverse STFT** : $x_w(m) = \frac{1}{2\pi} \int_{2\pi} X_w(e^{j\omega})e^{j\omega m}d\omega$
- Thus STFT is a 3D plot having $X(\omega, n)$



Deriving DFT Pair of Relations

- Frequency domain representation of DT signal $x(n)$
- Discrete time Fourier Transform (DTFT)
- **DTFT** : $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$
- **Inverse DTFT** : $x(n) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$
- Let $x(n)$ is a DT sequence block of N samples
- For time domain alias free condition, $N_s \geq N$ and let $N_s = N$
- Any frequency range of 2π say $-\pi$ to π is sampled into $N_s = N$ samples as $1/N_s$ will be interval between successive samples.
- Let $\omega = \frac{2\pi k}{N}$, where $k = 0, 1, \dots, (N-1)$



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- When sampled like this, ω becomes discrete.
- Discrete Fourier Transform (DFT)** : $X(k) = \sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi kn}{N}}$, where $k = 0, 1, \dots, (N - 1)$
- Accordingly, 2π maps to N and integration in inverse DTFT to summation over N interval.
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DFT Pair of Relations

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- Consider a finite duration sequence (FDS) $x(n)$ of N samples for frequency domain analysis
- Since DFT uses FDS, it can be used for computed spectrum for both stationary and non-stationary signals
- N is 10-30 ms duration of speech to ensure stationarity
- Speech is processed in blocks of 10-30 ms and for each block, DFT is computed to get frequency domain representation

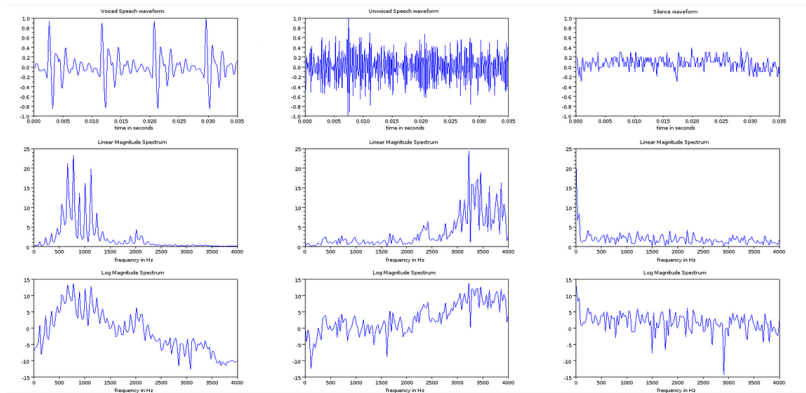


Linear vs Log Magnitude Spectrum

- In case of speech, large magnitude frequency components dominate in the linear magnitude spectrum.
- This gives a false picture of small magnitude frequency components.
- To reduce the dynamic range, the logarithmic version of the magnitude spectrum is taken.
- In case of log magnitude spectrum, both large and small magnitude frequency components are visible.

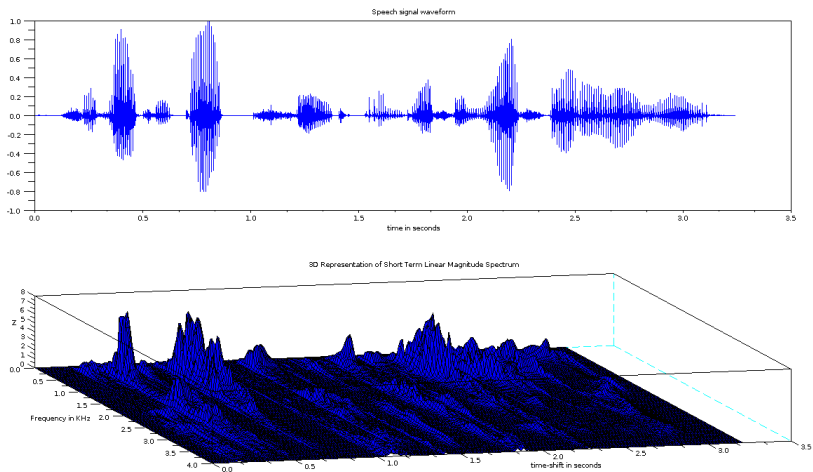


STFT of Voiced, Unvoiced and Silence Regions



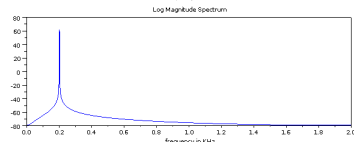
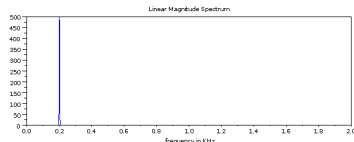
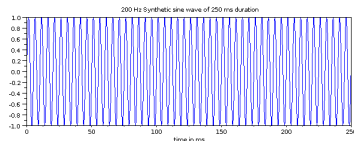
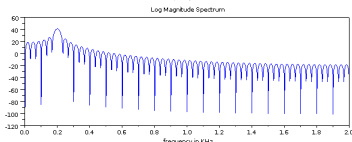
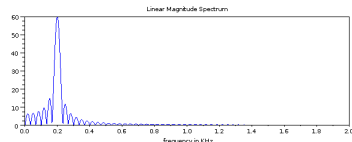
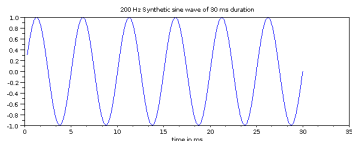
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Short Term Fourier Transform



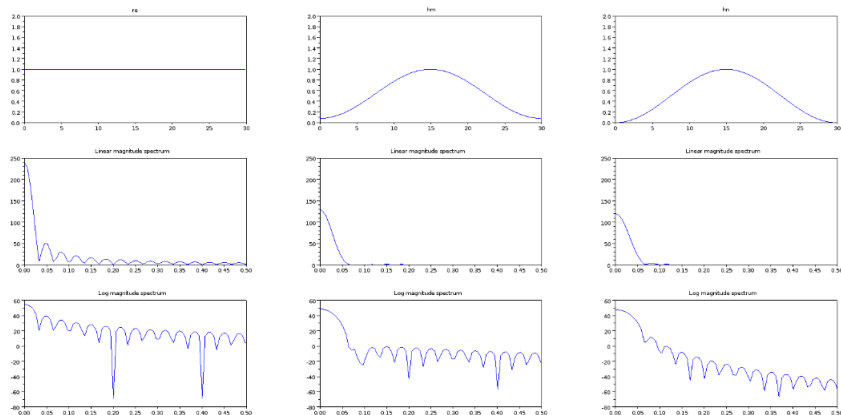
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True vs Convolved Spectra



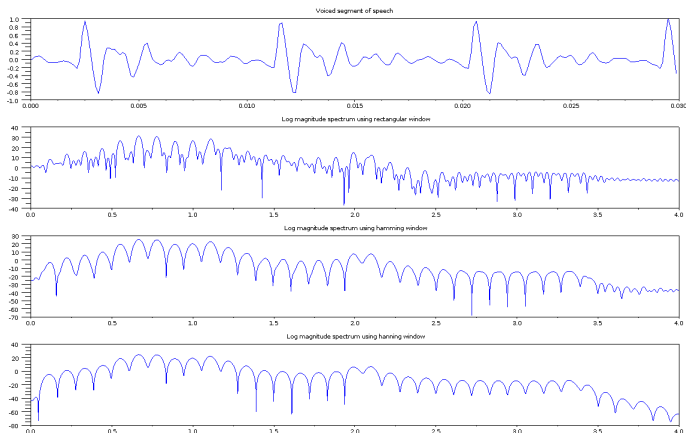
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Effect of Windowing Functions for STFT



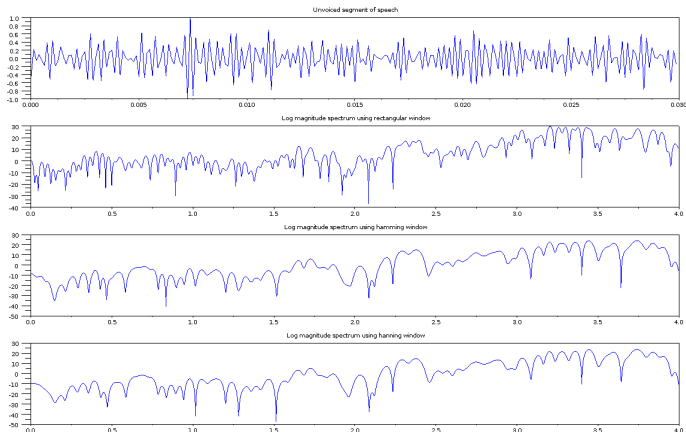
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Voiced Frame Spectra using Diff Window Fns



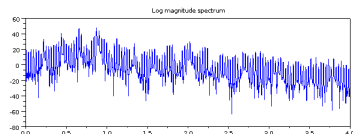
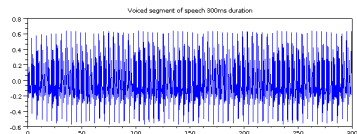
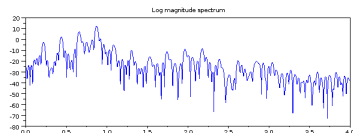
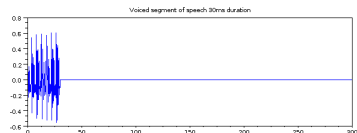
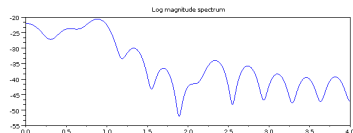
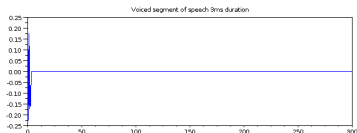
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Unvoiced Frame Spectra using Diff Window Fns



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Effect Window Size on STFT



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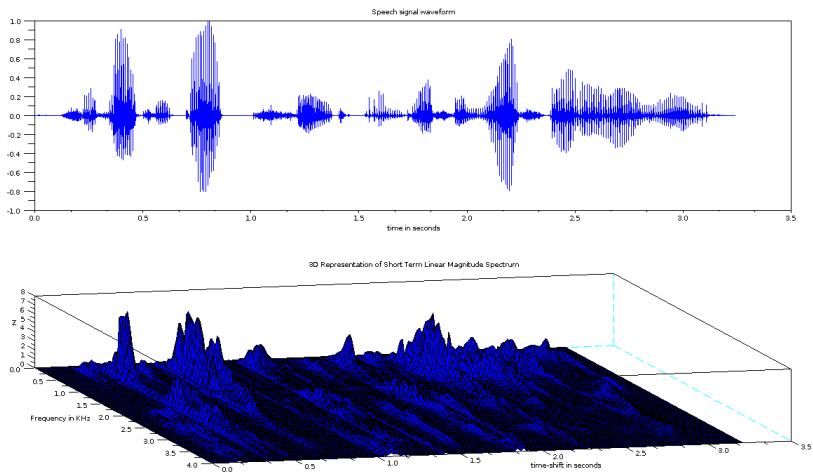


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Short Term Fourier Transform



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Spectrographic Representation

- **STFT** : $X(\omega, n) = \sum_{n=-\infty}^{\infty} x(n)e^{j\omega n}$
- **STFT** : $X(k, n) = \sum_{n=0}^{(N-1)} x(n)e^{j\frac{2\pi kn}{N}}, k = 0, 1, \dots, (N-1)$
- Convert given 3D plot of STFT into 2D representation.
- 3D amplitude vs time and frequency
- 2D amplitude as intensity vs time and frequency
- **Spectrogram** : 2D representation of 3D STFT magnitude information
- Also termed as visual or image representation of speech

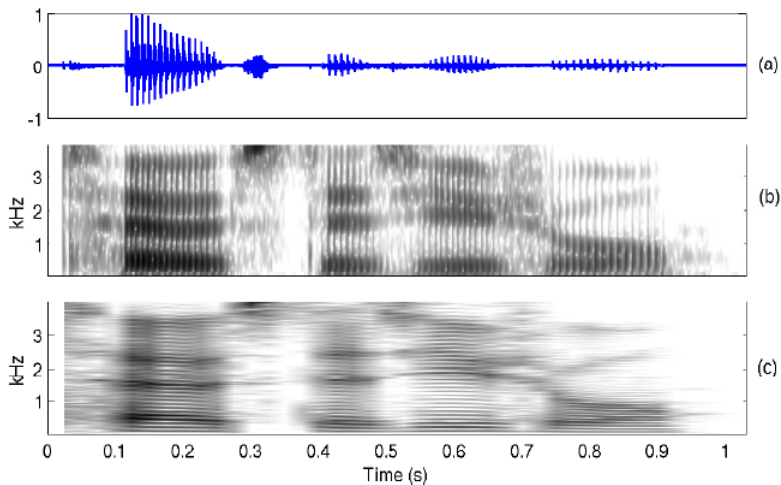


Construction of Spectrogram

- Entire frequency or spectral and time ranges are converted into time-frequency bins
- For instance, let the 4 kHz message frequency is converted into 20 bins, each bin of 200 Hz
- Let framesize is 20 ms and freameshift is also 20 ms
- Thus each time-frequency bin will be of 200 Hz- 20 ms size
- For a given frame of speech, compute STFT magnitude spectrum and find spectral magnitude in each tf bin
- Repeat the process for all the frames
- Convert spectral magnitudes into intensities by proportional mapping



Narrowband vs Wideband Spectrogram



- **Wideband** : Frequency band is big and time window small (5 ms)
- **Narrowband** : Frequency band is small and time window bigger (50 ms)

Need for Cepstral Analysis

- Speech signal is due to convolution of excitation source signal with vocal tract system impulse response
- Need for separating source and system components
- As both source and system components information not available, deconvolution is not possible
- Blind deconvolution is also not possible as there is no provision for applying different excitation signals
- Need a different approach than either convolution or deconvolution mentioned above



Cepstral Analysis

- We do source and system decomposition without deconvolution process
- Can be done using non-linear logarithmic operator property :
 $\log(ab) = \log a + \log b$
- Exploiting the property of Fourier transform
- Spreadout in one domain leads to localization in other domain
- Sinewave information in time domain vs frequency domain

Cepstral Analysis : Equations

- $X(j\omega) = \int x(t)e^{-2\pi ft} dt$
- $x(t) = \frac{1}{2\pi} \int X(j\omega)e^{2\pi ft} dt$
- $x(t) = \frac{1}{2\pi} \int |X(j\omega)|e^{\angle(X(j\omega))}e^{2\pi ft} dt$
- $x(t) = \frac{1}{2\pi} \int |E(j\omega)H(j\omega)|e^{\angle(E(j\omega)H(j\omega))}e^{2\pi ft} dt$
- $x_m(t) = \frac{1}{2\pi} \int |E(j\omega)H(j\omega)|e^{2\pi ft} dt$
- $x_c(t) = \frac{1}{2\pi} \int \log(|E(j\omega)H(j\omega)|)e^{2\pi ft} dt$
- $x_c(t)$ is called the cepstrum of $x(t)$